



PARSHWANATH CHARITABLE TRUST'S
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Department of Computer Science and Engineering
Data Science



ML-Based Transaction Pattern Analysis for Anti-Money Laundering

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Project Guide
Prof. Aishwarya Londhe

Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement
- System Design
- Technologies and methodologies
- Implementation
- Conclusion
- References

Introduction

- Anti-Money Laundering (AML) website utilizes machine learning to identify fraudulent activities, monitor financial transactions, and enhance regulatory compliance. With intelligent automation, it improves detection accuracy, minimizes risks, and ensures secure financial operations.

Motivation :

- Rising Need for Effective Anti-Money Laundering Solutions: Increasing complexity of financial crimes and the need for more robust systems to detect and prevent money laundering in real-time.
- Challenges with Current AML Systems: Traditional AML frameworks are often ineffective in handling sophisticated fraud techniques and high volumes of false positives, highlighting the need for smarter, adaptive AI-driven solutions.

Objectives

- Analyze financial data to detect suspicious transactions using isolation forest algorithm.
- Assign risk scores based on transaction history, demographics, and high-risk connections using random forest algorithm.
- Flag suspicious transactions.
- Create detailed Suspicious Activity Reports for flagged transactions.

Literature Survey of the existing system

Title	Author	Year	Outcomes	Methodology	Results
[1]Anti-money laundering systems: a systematic literature review	Alhanouf Abdulrahman Saleh Alsuwailam, Abdul Khader Jilani Saudagar	2020	Categorized AML techniques based on AI, data sources, and evaluation methods. Highlighted research gaps and suggested improvements for fraud detection models.	Conducted a systematic literature review of 27 documents from 2015 to 2020.	Provided insights into AI-driven AML models and identified challenges in real-world implementation.
[2]Network Analytics for Anti money laundering review and experimental evaluation	Bruno Deprez, Toon Vanderschueren, Bart Baesens, Tim Verdonck, Wouter Verbeke	2024	Analyzed network analytics techniques for AML, showing their effectiveness in detecting suspicious transactions. Found graph neural networks to be highly accurate.	Performed a systematic literature review and experimental evaluation using the Elliptic dataset.	Demonstrated that network-based models outperform traditional AML techniques in predictive accuracy.

Literature Survey of the existing system

Title	Author	Year	Outcomes	Methodology	Results
[3] A systematic review of anti-money laundering systems literature: Exploring the efficacy of machine learning and deep learning integration	Nadia Husnaningtyas, Ghalizha Failazufah Hanin, Totok Dewayanto, Muhammad Fahad Malik	2023	Evaluated AI-based AML approaches, emphasizing machine learning and deep learning. Found that hybrid AI models enhance fraud detection accuracy.	Conducted a systematic literature review of 26 studies published between 2018 and 2023.	Identified decision trees as commonly used and recommended combining AI techniques for better results.
[4] Anti-money laundering and anti-terrorism financing: a survey of the existing literature and a future research agenda	Noura Ahmed Al-Suwaidi, Haitham Nobanee	2020	Reviewed AML and anti-terrorism financing research, highlighting gaps in legal frameworks. Suggested stronger regulations and technological integration.	Performed a systematic literature review.	Emphasized the need for enhanced global cooperation and advanced AML technologies.

Limitations of existing systems

- Despite advancements in AI-driven Anti-Money Laundering (AML) systems, significant limitations persist, particularly with static risk assessment models and inadequate adaptation to evolving financial crime techniques.
- Performance Under Constraints: AML systems may struggle when dealing with limited transaction history or detecting unusual laundering methods, leading to false negatives (missed suspicious activities) or false positives (excessive alerts that require manual review).
- Complexity in Navigation: Financial analysts and compliance officers often struggle with navigating complex AML dashboards and reports, making it difficult to efficiently investigate suspicious transactions and take timely action.

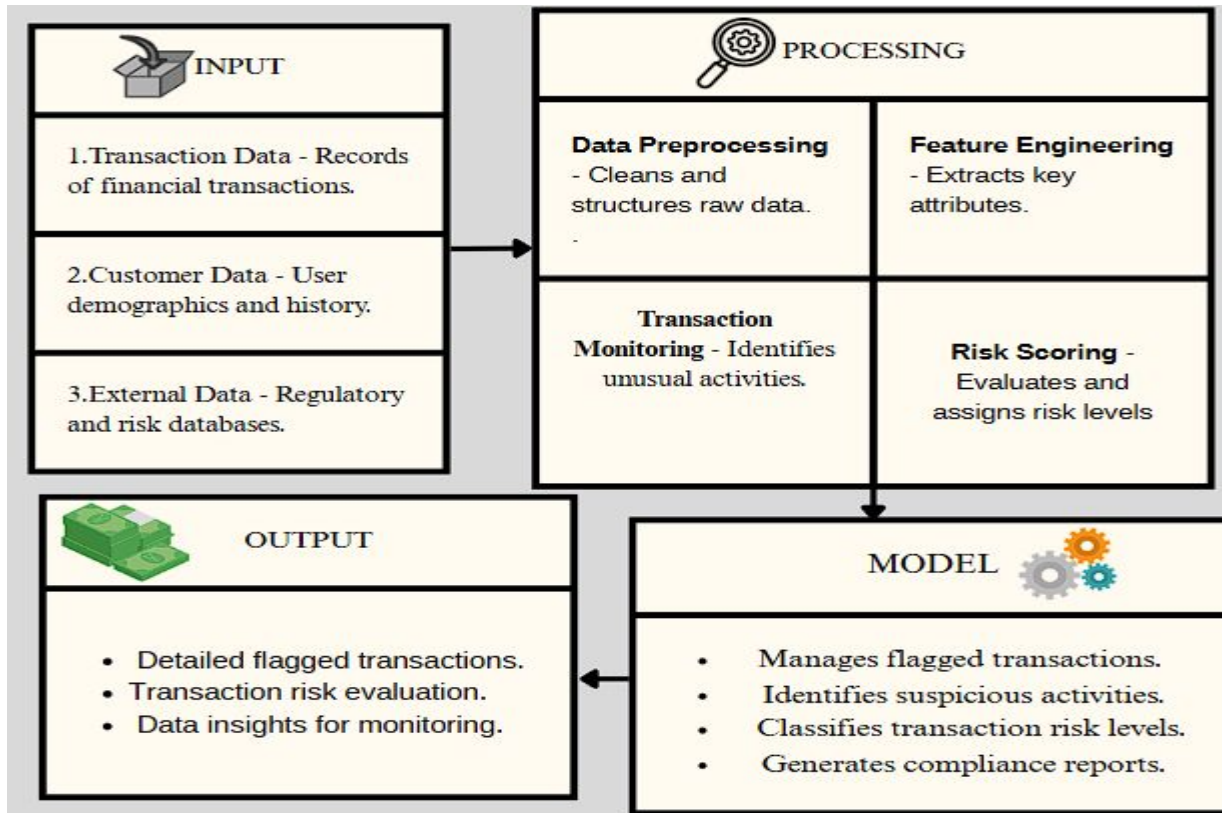
Problem statement

- Traditional AML systems fail to detect evolving fraud patterns and real-time suspicious activities. Excessive false positives overwhelm analysts, reducing focus on genuine threats. Generic risk assessments ignore specific customer behaviors, lowering detection accuracy. Reliance on historical data increases risk and limits adaptability.

Solution:

- Utilize Machine Learning Algorithms: Apply machine learning models to detect evolving fraud patterns, reduce false positives, and improve the accuracy of suspicious transaction detection.
- Personalized Risk Assessment and Detection: Implement customer-specific risk profiles and transaction monitoring to enhance detection capabilities and reduce reliance on historical data.

System Design



Technologies and methodologies

- **Web Technologies:**

Frontend: HTML5, CSS3, JavaScript

Backend: Flask

Database: PostgreSQL

- **Dataset:**

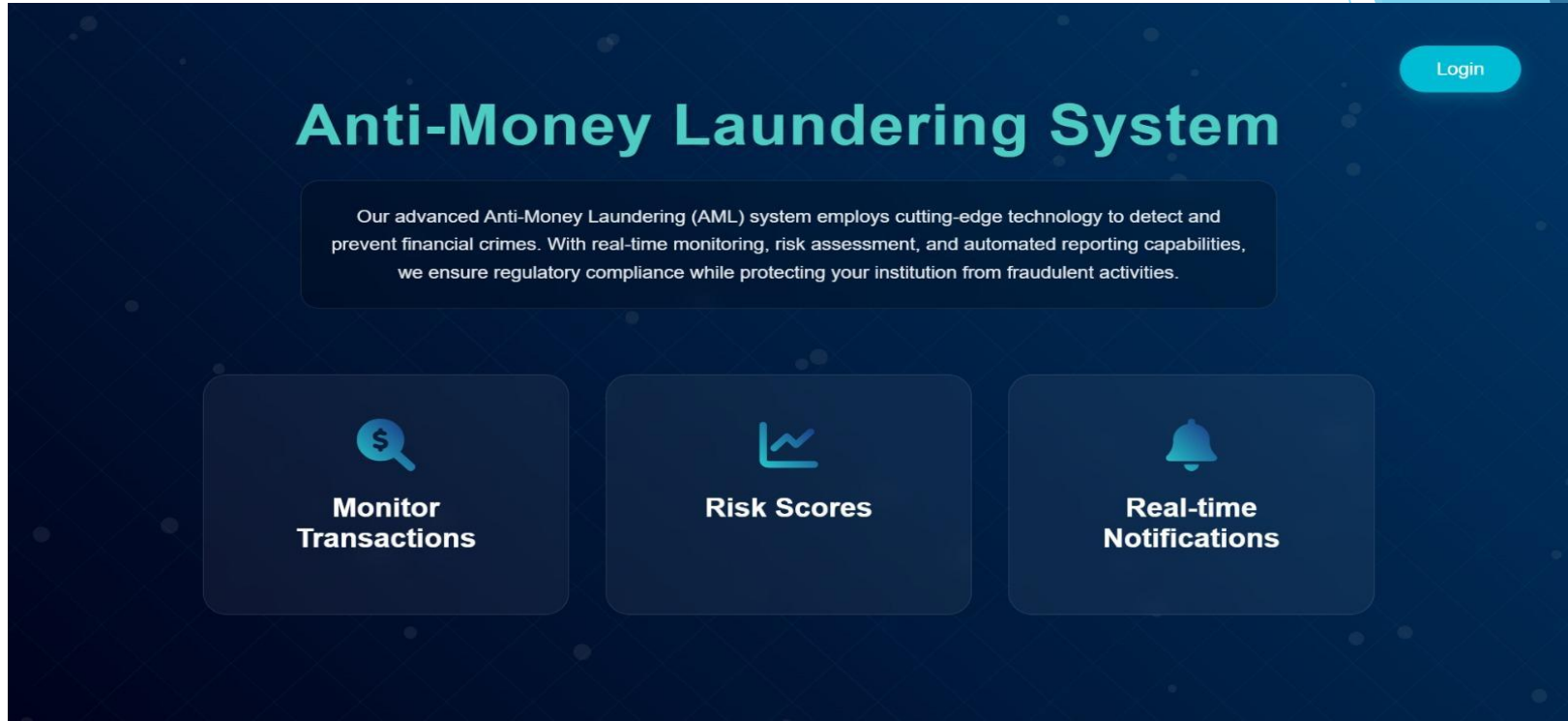
Synthetic Data (transactions and customer data)

- **Algorithms:**

Random forest (supervised)

Isolation forest(unsupervised)

Implementation



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Transaction Monitoring

Real-time monitoring and analysis of financial transactions

[Fetch New Transactions](#)

TRANSACTION ID	CUSTOMER ID	TIME	AMOUNT	TYPE	DESCRIPTION	STATUS
4560	826	Wed, 02 Apr 2025 01:28:00 GMT	\$35,237.06	deposit	Salary Deposit	⚠ Suspicious
103	20	Wed, 02 Apr 2025 19:21:00 GMT	\$33,545.21	payment	Online Purchase	⚠ Suspicious
2928	533	Wed, 02 Apr 2025 15:43:00 GMT	\$4,458.90	payment	Online Purchase	⚠ Suspicious
599	110	Wed, 02 Apr 2025 16:55:00 GMT	\$38,023.55	transfer	Cash Deposit	⚠ Suspicious
4305	785	Wed, 02 Apr 2025 01:26:00 GMT	\$10,293.75	withdrawal	Cash Withdrawal	⚠ Suspicious
4444	805	Wed, 02 Apr 2025 14:47:00 GMT	\$7,885.85	payment	Online Transfer	⚠ Suspicious
2342	430	Wed, 02 Apr 2025 10:44:00 GMT	\$20,560.35	withdrawal	Cash Withdrawal	⚠ Suspicious
108	21	Wed, 02 Apr 2025 19:46:00 GMT	\$45,234.09	deposit	Online Transfer	✅ Normal

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Risk Score Analysis

Analyze customer risk profiles and detect potential financial threats

cust1003

 Search

Customer Details

Name

Myra Chaudry

Age

61

Annual Income

\$3,874,884.61

Location

Rajpur Sonarpur, Sikkim

Email

zaraloke@singhal-keer.com

Logout

Anti-Money Laundering System

Our advanced Anti-Money Laundering System is designed to prevent financial crimes. With our state-of-the-art technology, we ensure regulatory compliance and

technology to detect and prevent suspicious activities. Our reporting capabilities, integrated with our advanced analytics, help identify and report on illicit activities.



**Monitor
Transactions**

Select Report Type - Google Chrome

127.0.0.1:5000/select-report

Select Report Type

Choose the format for your suspicious transactions report:

PDF Report

CSV Report



Generate Report

Conclusion

AI-based Anti-Money Laundering system plays a crucial role in enhancing financial security by detecting and preventing suspicious activities. By providing accurate and efficient monitoring, it helps organizations comply with regulations and reduce financial risks, ensuring a safer and more transparent financial environment.

References

- [1] Alhanouf Abdulrahman Saleh Alsuwailem, Abdul Khader Jilani Saudagar(2020)“Anti-money laundering systems: a systematic literature review” Provided insights into AI-driven AML models and identified challenges in real-world implementation.
- [2] Bruno Deprez, Toon Vanderschueren, Bart Baesens, Tim Verdonck, Wouter Verbeke(2024)“Network Analytics for Anti money laundering review and experimental evaluation” Demonstrated that network-based models outperform traditional AML techniques in predictive accuracy.
- [3] Nadia Husnaningtyas, Ghalizha Failazufah Hanin, Totok Dewayanto, Muhammad Fahad Malik(2023) “] A systematic review of anti-money laundering systems literature: Exploring the efficacy of machine learning and deep learning integration” Identified decision trees as commonly used and recommended combining AI techniques for better results.
- [4] Noura Ahmed Al-Suwaidi, Haitham Nobanee(2020) “Anti-money laundering and anti-terrorism financing: a survey of the existing literature and a future research agenda” Emphasized the need for enhanced global cooperation and advanced AML technologies.

Thank You...!!